



Figure 1: A diagram representation of the data centre set-up

On Debian and the likes, issue the following code:

```
# apt-get install ipvsadm keepalived
```

On Red Hat and the likes, use the following:

```
# yum install ipvsadm keepalived
```

The kernel module *ip_vs* and *ipip* may need to be loaded, but in my experience, these modules were automatically loaded when I used the *ipvsadm* command.

To start with, we will consider a scenario that has two data centres. There is one LVS load balancer in each data centre. For the sake of giving them names, we will call them *ipvsb11* and *ipvsb21*. Now we will configure the IPIP tunnel between the load balancers and the real servers—*rproxy1* and *rproxy2*, where *rproxy1* is in the first data centre and *rproxy2* is in the second.

Before we start the command configuration, have a look at Table 1 and Figure 1.

Please look at the following configuration carefully, and look at the host names in the prompt to identify to which server you need to type these commands:

```
rproxy1# modprobe ipip
rproxy1# ip addr add 192.168.1.214/32 dev tunl0
rproxy1# ip addr add 192.168.2.214/32 dev tunl1
rproxy1# ip link set tunl0 up arp off
rproxy1# ip link set tunl1 up arp off

rproxy2# modprobe ipip
rproxy2# ip addr add 192.168.1.214/32 dev tunl0
rproxy2# ip addr add 192.168.2.214/32 dev tunl1
rproxy2# ip link set tunl0 up arp off
rproxy2# ip link set tunl1 up arp off
```

Data Centre details

Data Centre	Host	Interface	IP Address	Role
Data Centre 1	ipvsb11	eth0	192.168.1.214/24	VIP
Data Centre 1	ipvsb11	eth1	192.168.1.13/24	DIP
Data Centre 2	ipvsb21	eth0	192.168.2.214/24	RIP
Data Centre 2	ipvsb21	eth1	192.168.2.13/24	DIP
Data Centre 1	rproxy1	eth0	192.168.1.14/24	RIP
Data Centre 1	rproxy1	tunl0 (no ARP)	192.168.1.214/32	VIP
Data Centre 1	rproxy1	tunl1 (no ARP)	192.168.2.214/32	VIP
Data Centre 2	rproxy2	eth0	192.168.2.2/24	RIP
Data Centre 2	rproxy2	tunl0 (no ARP)	192.168.1.214/32	VIP
Data Centre 2	rproxy2	tunl1 (no ARP)	192.168.2.214/32	VIP

Table 1

Now let us add two virtual servers—one to serve normal Web traffic and one for secure traffic on Port 443:

```
ipvsb11# ipvsadm -A -t 192.168.1.214:80 -s wlc
ipvsb11# ipvsadm -A -t 192.168.1.214:443 -s wlc
ipvsb11# ipvsadm -a -t 192.168.1.214:80 -r 192.168.1.14 -i
ipvsb11# ipvsadm -a -t 192.168.1.214:443 -r 192.168.1.14 -i
ipvsb11# ipvsadm -a -t 192.168.1.214:80 -r 192.168.2.2 -i
ipvsb11# ipvsadm -a -t 192.168.1.214:443 -r 192.168.2.2 -i

ipvsb21# ipvsadm -A -t 192.168.2.214:80 -s wlc
ipvsb21# ipvsadm -A -t 192.168.2.214:443 -s wlc
ipvsb21# ipvsadm -a -t 192.168.2.214:80 -r 192.168.1.14 -i
ipvsb21# ipvsadm -a -t 192.168.2.214:443 -r 192.168.1.14 -i
ipvsb21# ipvsadm -a -t 192.168.2.214:80 -r 192.168.2.2 -i
ipvsb21# ipvsadm -a -t 192.168.2.214:443 -r 192.168.2.2 -i
```

The various options used with *ipvsadm* are:

- *-A* adds a virtual server
- *-a* adds a real server
- *-t* specifies that the virtual server is a *tcp* service
- *-s* is a scheduling method
- *-r* represents the real server
- *-i* is an option which specifies that an IPIP tunnel is to be used. IPVS will generate the IPIP packets on its own

You can view the IPVS configuration using the *-L* option with the *ipvsadm* command. The *-n* option saves some DNS look-up time in case the real servers do not have a DNS or host file entry.

```
ipvsb11# ipvsadm -Ln
IP Virtual Server version 1.2.1 (size=4096)
Prot LocalAddress:Port Scheduler Flags
-> RemoteAddress:Port Forward Weight ActiveConn InActConn
TCP 192.168.1.214:https wlc persistent 86400
-> 192.168.1.14:https Tunnel 1 0 0
TCP 192.168.1.214:www rr
-> 192.168.1.14:www Tunnel 1 0 0
```

A similar output is produced at the *ipvsb21*.

Please make sure that the router at the Data Centre